Day 2 - Docker - 3 Tier App Deployment

- Created Dockerfile for backend

docker > day-2 > mern > backend > Dockerfile

```
FROM node:20-alpine

WORKDIR /app

COPY package*.json ./

RUN npm ci --omit=dev

COPY --chown=node:node . .

USER node

EXPOSE 5050

CMD ["node", "server.js"]
```

- Created Dockerfile for frontend using multi stage build

docker > day-2 > mern > frontend > Dockerfile

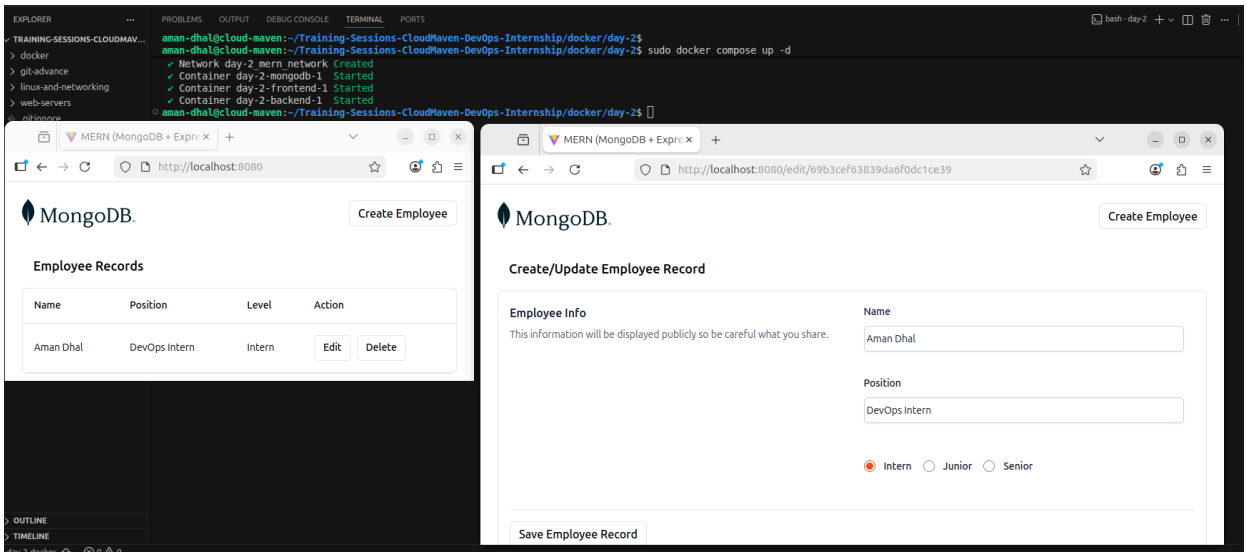
```docker
# -------- Stage 1: Build the React app --------
FROM node:20-alpine AS builder

# working directory
WORKDIR /app

# copy dependency files first (better caching)
COPY package*.json ./

# install dependencies
RUN npm ci

# copy source code
COPY . .

# build the app
RUN npm run build


# -------- Stage 2: Serve with Nginx --------
FROM nginx:alpine

# remove default nginx files
RUN rm -rf /usr/share/nginx/html/*

# copy built static files
COPY --from=builder /app/dist /usr/share/nginx/html


EXPOSE 80

# start nginx in foreground
CMD ["nginx", "-g", "daemon off;"]
```

- Created docker compose file to easily build and deploy all application components at once

docker > day-2 > docker-compose.yaml

```yaml
services:
  backend:
    build: ./mern/backend
    ports:
      - "5050:5050"
    networks:
      - mern_network
    environment:
      MONGO_URI: mongodb://mongodb:27017/mydatabase
    depends_on:
      - mongodb

  frontend:
    build: ./mern/frontend
    ports:
      - "8080:80"
    networks:
      - mern_network

  mongodb:
    image: mongo
    ports:
      - "27017:27017"
    networks:
      - mern_network
    volumes:
      - mongo-data:/data/db

networks:
  mern_network:
    driver: bridge

volumes:
  mongo-data:
    driver: local  # Persist MongoDB data locally
```

- Successfully deployed and tested mern stack app by executing docker compose up -d


EXPLORER
TRAINING-SESSIONS-CLOUDMAV...
docker
git-advance
linux-and-networking
web-servers
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
aman-dhal@cloud-maven:~/Training-Sessions-CloudMaven-DevOps-Internship/docker/day-2$
aman-dhal@cloud-maven:~/Training-Sessions-CloudMaven-DevOps-Internship/docker/day-2$ sudo docker compose up -d
✔ Network day-2_mern_network Created
✔ Container day-2-mongodb-1 Started
✔ Container day-2-frontend-1 Started
✔ Container day-2-backend-1 Started
aman-dhal@cloud-maven:~/Training-Sessions-CloudMaven-DevOps-Internship/docker/day-2$
MERN (MongoDB + Expre
http://localhost:8080
MongoDB.
Create Employee
Employee Records
Name
Position
Level
Action
Aman Dhal
DevOps Intern
Intern
Edit
Delete
http://localhost:8080/edit/69b3cef63839da6f0dc1ce39
Create/Update Employee Record
Employee Info
This information will be displayed publicly so be careful what you share.
Name
Aman Dhal
Position
DevOps Intern
Intern
Junior
Senior
Save Employee Record
OUTLINE
TIMELINE